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Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Time: 03 Hours

Code: 110601 ***Subject: Measurements and Instrumentation*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): **[14]**

(a) Which system analyzes brain wave activity?
(i) EOG (iii) ECG
(ii) EMG (iv) EEG

(b) Which type of sensor is used to measure flow?
(i) Potentiometer (iii) Hall effect sensor
(ii) Venturi tube (iv) Thermocouple

(c) What does "Six Sigma" focus on?
(i) Enhancing resolution (iii) Reducing defects in processes
(ii) Improving precision (iv) Minimizing systematic errors

(d) What is the mean of the following data set: 4, 6, 8, 10?
(i) 6 (iii) 8
(ii) 2 (iv) 9

(e) Which instrument is used for ultrasonic diagnostic measurement?
(i) Thermocouple (iii) Ultrasound transducer
(ii) Oscilloscope (iv) Megger

(f) What is an example of a non-invasive Cardiac output measurement diagnosis?
(i) Temperature measurement (iii) ECG
(ii) Blood pressure measurement (iv) All of the above

(g) What is the function of a transducer in bio-medical instruments?
(i) Convert electrical signals into light (iii) Measure pressure
(ii) Measure electric current (iv) Convert biological signals into electrical signals

(h) What does a Megger measure?
(i) Current (iii) Capacitance
(ii) Insulation resistance (iv) Voltage

(i) What does a digital multi-meter measure?
(i) Only current (iii) Only voltage
(ii) Only resistance (iv) Voltage, current, and resistance

(j) What type of instrument is used to measure inductance?
(i) Ohmmeter (iii) Ammeter
(ii) LCR meter (iv) Voltmeter

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Q.2 Solve both questions: **[14]**

(a) Discuss the use of instrument transformers for current and voltage measurement in power systems.

(b) Explain the working principle of Maxwell's bridge with phasor diagram.

Q.3 Solve both questions: **[14]**

(a) Explain the Hall effect and its application in Hall sensors for current and voltage measurements.

(b) Describe the working principle of a Digital Storage Oscilloscope (DSO).

Q.4 Solve both questions: **[14]**

(a) Define and differentiate between true value, accuracy, and precision in measurement systems.

(b) Explain the concept of True RMS measurement. Why is it important for accurate AC voltage and current measurements?

Q.5 Solve both questions: **[14]**

(a) Explain the Six-Sigma approach to quality control in industrial processes.

(b) Describe the working principles of temperature transducers thermocouples and RTD.

Q.6 Solve both questions: **[14]**

(a) Explain hysteresis and dead-band in measurement systems. How do they influence the performance of transducers?

(b) Explain how bio-electric signals like ECG, EEG, and EMG are generated and measured.

Q.7 Solve both questions: **[14]**

(a) Discuss the types of errors that occur in measurements. Explain the difference between systematic and random errors with practical examples.

(b) Discuss non-invasive diagnostic measurements used in healthcare.

Q.8 Solve both questions: **[14]**

(a) Describe the role of sensitivity in measurement systems. How is sensitivity calculated, and what are the consequences of low or high sensitivity in sensors?

(b) Explain the use of Meggers in insulation resistance testing. Why is insulation testing crucial in high-voltage systems?

Q.9 Write short notes on any two of the following: **[14]**

(a) CAT scan techniques

(b) Cardiovascular measurements

(c) Capacitance Measurement with phasor

(d) LVDT

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